

Executive Summary

Final supervisor-ready PhD proposal

CompressAI semantic utility-aware Earth Observation research platform

Word count: 4863

Executive Summary

Semantic UtilityAware Adaptive Communication for WildfireCentric Earth Observation Systems Under Resource Constraints

This PhD proposal investigates how Earth Observation systems can preserve wildfire relevant information when satellite communication resources are severely limited. The project is motivated by a practical problem in modern space systems: satellites can increasingly observe more than they can immediately transmit. This is especially important for wildfire monitoring, where the operational value of an image depends on whether fire, smoke, burn scars, and affected regions remain visible and detectable after compression.

The proposed research does not argue that established codecs such as JPEG, JPEG2000, or CCSDS image compression should be replaced. These methods remain strong, efficient, and operationally important. Instead, the proposal introduces a complementary high compression mode for situations where a satellite must prioritise mission relevant information under strict bandwidth, energy, or contact window constraints. The key question is whether semantic utility aware token prioritisation can preserve wildfire relevant information more effectively than non semantic token selection, and whether it can provide useful communication savings in Earth Observation scenarios.

The project builds on an existing research platform called CompressAI. The platform already supports wildfire utility estimation, VQVAE tokenisation, token ranking, adaptive token retention, reconstruction, Semantic Utility Score evaluation, statistical testing, and satellite communication analysis. This means the PhD starts from a working experimental foundation rather than from a speculative idea.

The central metric is the Semantic Utility Score (SUS), a 0100 mission oriented measure combining detector retention, object retention, relevance retention, and important region preservation. SUS is designed to answer a question that conventional image metrics do not directly answer: how much wildfire relevant information survives compression and reconstruction? Standard image metrics such as PSNR, SSIM, LPIPS, compression ratio, and bandwidth saved remain part of the evaluation, but SUS adds mission relevance.

Preliminary evidence has been generated across DFire, FLAME, and a 100 scene Sentinel2/CEMS benchmark. On DFire, utility aware token retention achieved $SUS = 83.14$, detector retention = 0.849, bandwidth saved = 95.10 percent, and compression ratio = 25.30x. On FLAME, the same operating point achieved $SUS = 65.84$, detector retention = 0.834, and bandwidth saved = 96.27 percent. On Sentinel2/CEMS, utility aware 45 percent token retention achieved $SUS = 79.52$, detector retention = 0.741, bandwidth saved = 99.35 percent, and compression ratio = 155.77x.

These results show promise but also reveal the central tradeoff. JPEG remains very strong for visual fidelity and detector retention in several regimes. Utility aware token prioritisation is most compelling under severe communication constraints, where the goal is not perfect image reconstruction but preservation of enough wildfire utility to support rapid interpretation. This honest comparison strengthens the proposal because it avoids overclaiming and defines the operating conditions under which the research contribution

matters.

The PhD research programme will expand the current evidence into a rigorous Earth Observation study. It will scale Sentinel2 validation beyond the existing 100scene benchmark, integrate FIRMS activefire detections and burnedarea masks, evaluate retention sweeps from 10 to 100 percent, compare against JPEG and learnedcompression baselines, and conduct statistical analysis using paired tests, Wilcoxon tests, Cohen's d, and bootstrap confidence intervals. The work will also evaluate communication savings and edgedeployment constraints relevant to CubeSatclass and onboard AI systems.

The expected contributions are fivefold. First, the project formalises SUS as a wildfirecentred semantic utility metric. Second, it develops utilityaware token prioritisation for learned image representations. Third, it validates the method on wildfire and Sentinel2 Earth Observation imagery. Fourth, it connects compression performance to satellite communication constraints. Fifth, it provides a reproducible benchmark framework suitable for IEEEstyle publication.

The proposal is well aligned with University of Luxembourg SnT, TU Delft, ESAaffiliated Earth Observation research, DLR, and European AIforspace programmes. It combines machine learning, remote sensing, semantic communication, and satellite systems engineering in a focused and experimentally testable PhD programme. The longterm vision is an Earth Observation system that can interpret observations onboard and transmit the most missionrelevant information first.

For a supervisor, the strongest aspect of the work is its specificity. Wildfire remains the primary mission throughout the proposal. Flood and shipdetection examples are not introduced as additional PhD themes; they remain possible validation tasks only. This keeps the scope realistic and preserves a coherent research line from utility definition to satellite communication analysis. The proposal also avoids the common weakness of semanticcommunication work in which meaning is discussed abstractly but not measured against a real mission. Here, the meaning of interest is wildfirerelevant evidence, and the measurement strategy is tied to detector retention, relevance mass, object preservation, and importantregion preservation.

The work is also feasible as a fouryear PhD because the experimental platform already exists. Year 1 can focus on consolidation and first publicationquality experiments rather than basic infrastructure. Year 2 can expand Sentinel2 and FIRMS validation. Year 3 can address edge deployment and communication modelling. Year 4 can consolidate the thesis, reproducibility package, and journal publications. The main risks are dataset quality, detector bias, and the strength of JPEG baselines. These risks are explicitly acknowledged and converted into evaluation questions rather than hidden weaknesses.

The proposal is suitable for European research environments because it links AI methodology with Earth Observation relevance and satellitesystem constraints. For SnT, the value lies in the connection between trustworthy AI systems, space applications, and communicationaware autonomy. For TU Delft, the work is relevant to remote sensing, geospatial AI, and spaceborne system design. For ESA and DLRoriented projects, the contribution is the demonstration of missionaware onboard prioritisation for datalimited Earth Observation. The

project is therefore not only a machinelearning compression study; it is a space AI research programme with a clear operational motivation.

The intended PhD outcome is a defensible scientific framework rather than a single prototype. The expected thesis contribution is to show when semantic utilityaware token transmission is useful, when conventional codecs remain preferable, and how wildfirecentric utility can be evaluated reproducibly under Earth Observation constraints. This balanced framing should make the proposal stronger for funded PhD applications, supervisor outreach, and later IEEE or arXiv publication.